Proof

Consider a plane whose perpendicular distance from the origin is p.

Let A be the foot of the perpendicular from O to the plane.

Let $\hat{d}$ be the unit normal vector in the direction of $\overrightarrow{OA}$.

Then $\overrightarrow{OA} = p\hat{d}$.

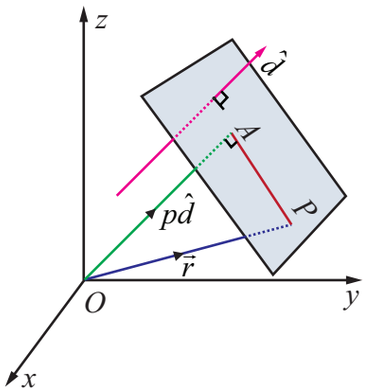


Fig. 6.24

If $\vec{r}$ is the position vector of an arbitrary point P on the plane, then $\overrightarrow{AP}$ is perpendicular to $\overrightarrow{OA}$.

Therefore, $\overrightarrow{AP} \cdot \overrightarrow{OA} = 0 \Rightarrow (\vec{r} - p\hat{d}) \cdot p\hat{d} = 0$

$\Rightarrow (\vec{r} - p\hat{d}) \cdot \hat{d} = 0$

which gives $\vec{r} \cdot \hat{d} = p$. ... (1)

The above equation is called the vector equation of the plane in **normal form**.

(b) Cartesian equation of a plane in normal form

Let l, m, n be the direction cosines of $\hat{d}$. Then we have $\hat{d} = l\hat{i} + m\hat{j} + n\hat{k}$.

Thus, equation (1) becomes

$$\vec{r} \cdot (l\hat{i} + m\hat{j} + n\hat{k}) = p$$

If P is (x,y,z), then $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Therefore, $(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (l\hat{i} + m\hat{j} + n\hat{k}) = p$ or $lx + my + nz = p$ ... (2)

Equation (2) is called the Cartesian equation of the plane in **normal form**.

Remark

(i) If the plane passes through the origin, then $p = 0$. So, the equation of the plane is $lx + my + nz = 0$.

(ii) If $\vec{d}$ is normal vector to the plane, then $\hat{d} = \dfrac{\vec{d}}{|\vec{d}|}$ is a unit normal to the plane. So, the vector equation of the plane is $\vec{r} \cdot \dfrac{\vec{d}}{|\vec{d}|} = p$ or $\vec{r} \cdot \vec{d} = q$, where $q = p|\vec{d}|$. The equation $\vec{r} \cdot \vec{d} = q$ is the vector equation of a plane in **standard form.**

Note

In the standard form $\vec{r} \cdot \vec{d} = q$, $\vec{d}$ need not be a unit normal and q need not be the perpendicular distance.

6.8.2 Equation of a plane perpendicular to a vector and passing through a given point

(a) Vector form of equation

Consider a plane passing through a point A with position vector $\vec{a}$ and $\vec{n}$ is a normal vector to the given plane.

Let $\vec{r}$ be the position vector of an arbitrary point P on the plane.

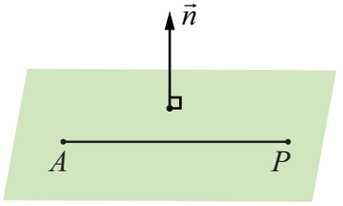


Fig. 6.25

12th_Maths_EM_Vol1_CH 6 Vector Algebra.indd 256 16-12-2022 14:25:30